

Statistics for Data Science

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References

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Introduction

Def. (Ω, K, P) is called a probability space, where:

- 1) Ω – sample space = the set of all possible outcomes
- 2) $K \subset P(\Omega)$ the power set of Ω (the set of all subsets of Ω)

$$K \text{ is a } \sigma\text{-algebra} \begin{cases} \emptyset \in K \\ A \in K \Rightarrow \bar{A} \in K \\ A_n \in K \Rightarrow \bigcup_{n=1}^{\infty} A_n \in K \end{cases}$$

- 3) $P: K \rightarrow [0,1]$ is a probability measure function $P(\Omega) = 1$

A measure function:

$$\begin{cases} P : K \rightarrow [0, \infty] \\ P(\emptyset) = 0 \\ P(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} P(A_n) \text{ if } A_n \in K, A_i \cap A_j = \emptyset, \forall i \neq j \end{cases}$$

The elements of K are called random events.

Introduction

Def. If S is a set, the σ -algebra generated by S , written $B(S)$ is:

$$B(S) \stackrel{\text{def}}{=} \bigcap_{\substack{F \supset S \\ F \text{ is } \sigma\text{-algebra}}} F.$$

If $S = \mathbb{R}$ then $\rho = B(S) = B((a, b) \mid a, b \in \mathbb{R})$.

If S is finite or countable then $\rho = P(S)$.

(S, ρ) is a measurable space.

Def. $X: (\Omega, K) \rightarrow (S, \rho)$ is a random variable if $\forall B \in \rho \Rightarrow X^{-1}(B) \in K$.

$$P \circ X^{-1} : \rho \rightarrow [0,1], \quad (P \circ X^{-1})(B) = P(X^{-1}(B)).$$

Introduction

The cumulative distribution function (CDF) of X is:

$$F(x) \stackrel{\text{def}}{=} (P \circ X^{-1})(-\infty, x) = P(X < x)$$

$$F: \mathbb{R} \rightarrow [0, 1]$$

$$F(-\infty) = 0, F(\infty) = 1$$

F is monotonically increasing.

- For X a continuous random variable, $P \circ X^{-1}$ is specified by the density function $f(x)$, $f(x) \geq 0, \forall x$

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$f(x) = F'(x)$$

$$F(x) = \int_{-\infty}^x f(t) dt$$

Introduction

- For X a discrete random variable, $P \circ X^{-1}$ is specified by

$$\{p(x), x \in S\}, p(x) \geq 0, \forall x$$
$$\sum_{x \in S} p(x) = 1$$

$$X : \begin{pmatrix} x_1 & \cdots & x_n \\ p_1 & \cdots & p_n \end{pmatrix}$$
$$F(x) = \sum_{x_i < x} P(X = x_i)$$

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx \text{ (or } \sum_{i=1}^n x_i p_i \text{ for the discrete case)}$$

$$Var(X) = E((X - E(X))^2) = E(X^2) - E(X)^2$$

$$\sigma = \sqrt{Var(X)} - \text{standard deviation}$$

Stochastic models of different phenomena of interest

- Categorical random variables/vectors.
- Quantitative discrete/continuous random variables/vectors.

Def. A stochastic process is a family of random variables $\{X_t, t \in T\}$ defined on a probability space (Ω, K, P) .

time T , states S :

time: $\mathbb{N}/\mathbb{Z}/\mathbb{R}/[0, \infty)$
state: discrete/ $\mathbb{R}/\mathbb{R}^d$

Stochastic models of different phenomena of interest

- Observations in “transversal” studies (at a fixed moment of time)
 $\{X_1, \dots, X_n\}$ random variables, independent and identically distributed (i.i.d.) like the stochastic model $X: \Omega \rightarrow S$.
Statistical data are the observed values $(x_1, \dots, x_n) \in S^n$.
- Observations in “longitudinal” studies (on a fixed time interval)
 $\{X_t, t \leq t_n\} \subset \{X_t, t \in T\}$
Statistical data are the observed part of the process trajectory
 $(x_t, t \leq t_n)$

Stochastic models of different phenomena of interest

Parametric statistics

Hypothesis: statistical data come from a stochastic model whose distribution has a known functional form, but it depends on an unknown parameter $\theta \in \mathbb{R}^k, k \geq 1$.

- Statistical data are available;
- Looking for estimators of θ or tests on hypotheses over θ .

Stochastic models of different phenomena of interest

Nonparametric statistics – makes minimal assumptions about the distribution of the data:

- include methods that are distribution-free: e.g., the two sample Kolmogorov Smirnov test; or methods where the theoretical distribution is specified, including the parameters values: e.g., the one sample Kolmogorov Smirnov test;

- include techniques that don't assume a fixed model structure: e.g., nonparametric regression.

Example of stochastic models

1. Categorical 1- dim random variable:

X = satisfaction degree

$$S = \{vun, un, sa, vsa\}$$

$$P \circ X^{-1}: \begin{pmatrix} vun & un & sa & vsa \\ p_1 & p_2 & p_3 & p_4 \end{pmatrix} \quad p_i \geq 0, \sum_{i=1}^4 p_i = 1$$

2. Categorical 2 - dim random vector:

$X = (X_1, X_2)' = (\text{intention to vote, satisfaction degree about the current situation})$

$$S = \{(yes, vun), (yes, un), (yes, sa), (yes, vsa), \\ (no, vun), (no, un), (no, sa), (no, vsa)\}$$

Example of stochastic models

Quantitative discrete random variables:

3. Discrete uniform distribution – e.g., rolling a dice

$$X \sim U\{1, \dots, r\}, r \in \mathbb{N}, r \geq 2$$

$$S = \{1, 2, \dots, r\}$$

$$P(X = x) = \frac{1}{r}, x = 1, 2, \dots, r$$

$$E(X) = \frac{r + 1}{2}$$

$$\text{Var}(X) = \frac{r^2 - 1}{12}$$

Example of stochastic models

4. Bernoulli distribution – e.g., having “success” in a trial with two possible outcomes: **success/failure** (1/0)

$$X \sim B(1, \theta), \theta \in (0, 1), S = \{0, 1\}$$

$$P_{\theta}(X = x) = \theta^x (1 - \theta)^{1-x}, x = 0, 1$$

$$X: \begin{pmatrix} 0 & 1 \\ 1 - \theta & \theta \end{pmatrix}$$

$$E(X) = \theta$$

$$Var(X) = \theta(1 - \theta)$$

Example of stochastic models

5. Binomial distribution – e.g., number of “successes” in r independent trials with two possible outcomes: success/failure.

$$X \sim Bi(r, \theta), \theta \in (0, 1), S = \{0, 1, \dots, r\}$$

$$P_{\theta}(X = x) = C_r^x \theta^x (1 - \theta)^{r-x}, x = 0, 1, \dots, r$$

$$E(X) = r \cdot \theta$$

$$Var(X) = r \cdot \theta(1 - \theta)$$

Example of stochastic models

6. *Poisson* distribution – e.g., number of defective devices in a very large volume lot.

$$X \sim P_{\theta}(\theta), \theta \in (0, \infty), S = \mathbb{N}$$

$$P_{\theta}(X = x) = \frac{\theta^x}{x!} e^{-\theta}, x = 0, 1, \dots$$

$$E(X) = \theta$$

$$\text{Var}(X) = \theta$$

Example of stochastic models

7. Geometric distribution – e.g., the moment of the first “success” in a sequence of independent trials with two possible outcomes: success/failure.

$$X \sim Ge(\theta), \theta \in (0, 1), S = \mathbb{N}^*$$

$$P_{\theta}(X = x) = (1 - \theta)^{x-1} \cdot \theta, x = 1, 2 \dots$$

$$E(X) = \frac{1}{\theta}$$

$$Var(X) = \frac{1 - \theta}{\theta^2}$$

Example of stochastic models

Quantitative discrete random vectors

8. Multinomial distribution – e.g., sampling with replacement r – extractions from a ballot box containing balls of d colors

$$X = (X_1, \dots, X_d)' \sim M(r; p_1, \dots, p_d)$$

$$P_\theta(X = (x_1, \dots, x_d)') = \frac{r!}{x_1! \dots x_d!} p_1^{x_1} \dots p_d^{x_d}, \text{ where:}$$

$$S = \{(x_1, \dots, x_d)' \mid x_i \in \{0, 1, \dots, r\} \forall i = \overline{1, d}, \sum_{i=1}^d x_i = r\}$$

$$\theta = (p_1, \dots, p_d), p_i \in [0, 1] \forall i = \overline{1, d}, \sum_{i=1}^d p_i = 1$$

$$E(X_i) = rp_i, i = \overline{1, d}$$

$$Var(X_i) = rp_i(1 - p_i), i = \overline{1, d}$$

$$Cov(X_i, X_j) = -rp_i p_j, i \neq j$$

Example of stochastic models

Quantitative continuous random variables

9. Continuous uniform distribution – e.g., throw a dart at random in the interval $[0, \theta]$ – all the points are equally likely to be hit.

$$X \sim U(\theta), \theta \in (0, \infty)$$

$$f(x; \theta) = \begin{cases} \frac{1}{\theta}, & x \in [0, \theta] \\ 0, & x \notin [0, \theta] \end{cases}$$

$$E(X) = \frac{\theta}{2}$$

$$Var(X) = \frac{\theta^2}{12}$$

Example of stochastic models

10. Exponential distribution – e.g., the simplest model for lifetime.

$$X \sim \text{Expo}(\theta), \theta \in (0, \infty)$$

$$f(x; \theta) = \begin{cases} \frac{1}{\theta} \exp\left(-\frac{x}{\theta}\right), & x \in [0, \infty) \\ 0, & x \in (-\infty, 0) \end{cases}$$

$$E(X) = \theta$$

$$\text{Var}(X) = \theta^2$$

Example of stochastic models

11. Gama distribution – e.g., general model for lifetime.

$$X \sim \text{Gama}(\alpha, \theta), \alpha \in (0, \infty), \theta \in (0, \infty)$$

$$f(x; \alpha, \theta) = \begin{cases} \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} \exp\left(-\frac{x}{\theta}\right), & x \in [0, \infty) \\ 0, & x \in (-\infty, 0) \end{cases}, \text{ where:}$$

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$$

$$E(X) = \alpha\theta$$

$$\text{Var}(X) = \alpha\theta^2$$

Example of stochastic models

12. Normal (Gaussian) distribution.

$$X \sim N(\mu, \sigma^2), \theta \in (\mu, \sigma^2) \in \mathbb{R} \times (0, \infty)$$

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2} (x - \mu)^2\right), x \in \mathbb{R}$$

$$E(X) = \mu$$

$$Var(X) = \sigma^2$$

Example of stochastic models

Quantitative continuous random vectors

13. Normal distribution $N(d; O, I)$

- is the product of d normal distributions $N(0,1)$

$$X = (X_1, \dots, X_d)' \sim N(d; O, I)$$
$$f(X) = \frac{1}{(2\pi)^{\frac{d}{2}}} \exp \left\{ -\frac{1}{2} X' X \right\}, X = (x_1, \dots, x_d)' \in \mathbb{R}^d$$

Example of stochastic models

Quantitative continuous random vectors

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14. Normal distribution $N(d; \mu, \Sigma)$

$$X = (X_1, \dots, X_d)' \sim N(d; \mu, \Sigma)$$
$$f(X; \mu, \Sigma) = \frac{1}{(2\pi)^{\frac{d}{2}} (\det \Sigma)^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2} (X - \mu)' \Sigma^{-1} (X - \mu) \right\},$$
$$X = (x_1, \dots, x_d)' \in \mathbb{R}^d$$

$$E(X) = \mu$$

$$\text{Cov}(X, X') = E((X - E(X))(X - E(X))') = \Sigma$$

Σ is symmetric and positive definite matrix.

To do:

Write algorithms to generate the following random variables/vectors:

$$1) X: \begin{pmatrix} q_1 & \dots & q_m \\ p_1 & \dots & p_m \end{pmatrix}, \quad \sum_{i=1}^m p_i = 1, \quad p_i \geq 0$$

$$2) X \sim \text{Ge}(\theta), \theta \in (0,1)$$

$$3) X = (X_1, \dots, X_d)' \sim M(r; p_1, \dots, p_d), \sum_{i=1}^d p_i = 1, p_i \geq 0$$